SERRE VANISHING ON AFFINES (RETAINED WITH THE AUTHOR'S SLOGAN BLOCK): FOR A QUASI-COHERENT MODULE F ON A SCHEME X AND ANY AFFINE OPEN U, THE COHOMOLOGY H^P(U, F) VANISHES FOR ALL P > 0.

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Editorial note. Statement span is lines 362-372 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 374-391, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33f.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01EW: sources/stacks-coherent/pinned-original-source.tex, source lines 391–391. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01X9: sources/stacks-coherent/pinned-original-source.tex, source lines 262–269. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01HS: sources/stacks-coherent/pinned-original-source.tex, source lines 391–391. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. For any affine open $U \subset X$ we have $H^p(U, \mathcal{F}) = 0$ for all $p > 0$.*

Proof. We are going to apply Cohomology, Lemma 01EW. As our basis $\mathcal{B}$ for the topology of X we are going to use the affine opens of X. As our set Cov of open coverings we are going to use the standard open coverings of affine opens of X. Next we check that conditions (1), (2) and (3) of Cohomology, Lemma 01EW hold. Note that the intersection of standard opens in an affine is another standard open. Hence property (1) holds. The coverings form a cofinal system of open coverings of

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any element of $\mathcal{B}$, see Schemes, Lemma 01HS. Hence (2) holds. Finally, condition (3) of the lemma follows from Lemma 01X9. $\square$